CHINA / USA / AI

The new technology borders

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AI

05

BIO

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SOLAR

The new technology borders
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A visual guide to China's Announcement 48, America's AI controls, and what comes next for open models.



Artificial intelligence 23 / Biomedicine 5 / Solar energy 12 / CN48

Research edition: 2 November 2026. We separate rules already operating, proposals still open for comment, and our own analysis. This is a public explanation, not a ruling on a particular transaction.

Two countries. Different gates.

CHINA | Registration now, catalogue changes proposed. The 16 October announcement opens consultation on 40 technologies: 23 in AI, 5 in biomedicine and 12 in solar. Export registration starts immediately for covered transfers, including unfinished ones. The proposed list contains 14 prohibited and 26 restricted technologies. The comment period ends on 15 November 2026. CN48 CNQA

UNITED STATES | Chip controls operate, model-rule status needs care. Advanced AI chips face license requirements and checks on their destination, buyer and use. Since 15 January 2026, qualifying H200-class exports from the US to China or Macau can receive individual review under conditions. That is permission to apply, not automatic permission to ship. The AI Diffusion Rule remains in the current published rules, while BIS has announced non-enforcement. GAO challenged the procedure for that policy in May 2026. US26 EAR742 USNE GAO26 EAR774

The central finding

The announcement does not establish a blanket ban on Chinese open models. Already public AI technology is excluded in the draft. New open-source exports of restricted AI technology are offered automatic licensing or similar facilitation. An exclusion and a permission procedure are different things. Neither erases other laws. CN48 CN-A1 CNQA

Read the evidence, then the possibilities

We start with what changed, walk through all 40 technology points, explain AI and open releases, and compare the US system. Each chapter links to evidence. The interactive explorer helps us find an item by sector or status. The scenario guide explains examples; it does not decide legal eligibility.

The two official attachments are available as original WPS files and locally converted PDFs. The converted PDFs are reading copies, not documents issued as PDFs by MOFCOM. The local file named china_export_control.pdf is an unrelated organoid presentation and is not used as evidence.

Official documents

- **48 / 1** Original WPS | Converted PDF

- **48 / 2** Original WPS | Converted PDF

48 / HTML / PDF

01. What changed, and when

One announcement, two clocks



16 OCT 2026 / Registration starts / 15 NOV 2026 / Comments close / DATE NOT FIXED / Final catalogue takes effect / CN48

MOFCOM and the Ministry of Science and Technology issued Announcement 48 on 16 October 2026. The first clock concerns a proposed change to the Catalogue of Technologies Prohibited or Restricted from Export. The second concerns temporary registration before covered exports. Registration began on publication. The proposed prohibitions and licensing categories have not become final merely because the registration duty has started. CN48, sections I and III

The consultation runs from 16 October through 15 November 2026. The temporary registration period ends when the revised catalogue is formally issued and implemented. The announcement gives no fixed calendar date for that event. We found no final replacement in the sources checked for this edition. A later announcement could change this account. CN48, sections II and III

What counts as a transfer

Registration covers transferring ownership abroad, licensing domestic technology for overseas use, and a domestic technology owner or actual controller becoming an overseas entity. MOFCOM may specify further situations by notice. Exports already started but not completed are included. A company transaction can therefore matter even when no box crosses a border. That last sentence is our reading of the ownership and control provision. CN48, section III(2)

The relevant application is the technology trade information service on MOFCOM's unified platform at ecomp.mofcom.gov.cn. Registration collects exporter and recipient information, destination, technology, use and expected value. Several technologies going to the same recipient in one export may be combined. Other cases should normally be filed separately. Registration is not itself an export license or proof that another restriction has been satisfied. CN48, section III CN-A2

Why the wording matters

A **prohibition** would stop a covered export within its stated scope. A **restriction** would create a permission route. **Automatic licensing** still describes a procedure. An **exclusion** removes a specified situation from that particular scope. We must read each qualification before using the headline category. CN-A1, notes 1 to 4 CNQA

MOFCOM describes the revision as routine technology-trade management, not a measure aimed at a particular country, region or entity. That is the government's stated purpose. It is not evidence that the commercial effects will be the same in every market. CNQA, background question

A worked example

Suppose a company plans to license private model technology to an overseas partner in November. We first identify the exact technology and the relevant proposed entry. We then check whether the transfer falls within the temporary registration scope and existing law. We cannot simply wait for the consultation to end. Conversely, the draft's proposed prohibited status is not enough to claim that every such export is already banned by Announcement 48. This is an illustrative application of the two clocks, not a transaction clearance. CN48 CN-A1

02. The 40-technology atlas

Count the technologies, not the table rows

The official headline counts 40 technologies. Attachment 1 groups several technology points under the same catalogue code. It does not contain 40 separate code rows. The explorer below splits those points into 40 reading entries and retains their original code and point number. This is an editorial index, not a replacement legal list. CN48 CN-A1

AI accounts for 23 points, with 6 proposed prohibited points and 17 restricted points. Biomedicine has 3 prohibited and 2 restricted points. Solar has 5 prohibited and 7 restricted points. Those add to the official totals of 14 and 26. The sector breakdown by status is our count of Attachment 1's numbered points. CN-A1 CNQA

How to use the atlas

We read the short description to understand the subject, then open the attachment at the stated code. A number alone is rarely sufficient. Some entries require every condition, some require any one condition, and some apply only to a special purpose or setting. The summaries preserve the main boundary but do not reproduce every measurement condition. The original Chinese text governs.

Public-domain AI exclusions and the footnotes on easier licensing apply across relevant entries. Solar also covers specified equipment that carries the technology. Biomedicine distinguishes reusable platforms from work on individual drugs. These notes are part of the substance, not optional small print. CN-A1, notes 1 to 4

01. Sensitive deepfake detection

266401J / 1 | Artificial intelligence | Proposed prohibited

Detection technology mainly for China's cross-border network-data security, with both accuracy and recall at least 95%. The specified purpose is part of the boundary. CN-A1

02. AI-driven malicious code

266402J / 1 | Artificial intelligence | Proposed prohibited

Specified malicious-code generation or mutation usable against highly sensitive systems or national defenses, meeting one of the listed attack-success tests. This is not every coding assistant. CN-A1

03. Adaptive malicious behavior

266402J / 2 | Artificial intelligence | Proposed prohibited
Specified technology for adapting in adversarial environments or changing behavior while running, with listed evasion or rewriting properties and the sensitive-use qualifier. CN-A1

04. Automated malicious variants

266402J / 3 | Artificial intelligence | Proposed prohibited
Specified automated production and distributed deployment of evading malicious variants, subject to listed scale or performance tests and the sensitive-use qualifier. CN-A1

05. Models trained on core data

266501J / 1 | Artificial intelligence | Proposed prohibited
Large models trained using China's legally defined core data and then provided as services. The data category is determined by law, not by a model's brand. CN-A1

06. Agents for highly sensitive settings

266502J / 1 | Artificial intelligence | Proposed prohibited
Agents mainly for critical infrastructure, defense or core government settings that also reach the performance criteria referenced in 266505X. CN-A1

07. High-accuracy deepfake detection

266402X / 8 | Artificial intelligence | Proposed restricted
Deepfake detection systems and equipment-manufacturing technology with both accuracy and recall at least 98%. This adds a point to an existing category. CN-A1

08. Hidden information in images and video

266402X / 9 | Artificial intelligence | Proposed restricted
Embedding and extracting hidden information meeting all three groups of tests: robustness, false positives at most 0.1%, and near-imperceptibility. The attachment gives alternative metrics within groups. CN-A1

09. AI-assisted vulnerability discovery

266404X / 6 | Artificial intelligence | Proposed restricted
Tools meeting any of six listed capability tests, including benchmark scores, independent exploitation capability or specified efficiency gains. Ordinary security testing is not automatically covered. CN-A1

10. Voiceprint recognition

266501X / 19 | Artificial intelligence | Proposed restricted
Speaker-identification technology meeting any listed recognition, response-speed, anti-spoofing or minority-language/dialect criterion. Each test has its own measurement conditions. CN-A1

11. Chinese and minority-language translation

266501X / 20 | Artificial intelligence | Proposed restricted
Technology mainly for Chinese and minority languages meeting a listed human-rating or semantic-fidelity and BLEU test. Language purpose and test conditions matter. CN-A1

12. Models trained on important data

266504X / 1 | Artificial intelligence | Proposed restricted
Large models trained on China's legally defined important data and then provided as services, with qualified exclusions for trace use with safeguards or lawfully public data. CN-A1

13. Advanced large-model core technology

266504X / 2 | Artificial intelligence | Proposed restricted
Core model technology if any test is met: at least 10^26 training calculations, at least 2 trillion parameters, or a qualifying global top-10 ranking at export time. Includes weights, checkpoints, code and architecture. CN-A1

14. Agent planning

266505X / 1 | Artificial intelligence | Proposed restricted
Natural-language instructions split into at least 20 executable subtasks, with decomposition accuracy at least 80% and average subtask success at least 90%. All these conditions apply. CN-A1

15. Many agents working together

266505X / 2 | Artificial intelligence | Proposed restricted
At least 60 concurrent agents, task time at most twice the minimum single-agent time for the same effect, and final-answer agreement at least 85%. CN-A1

16. Tool-using agents

266505X / 3 | Artificial intelligence | Proposed restricted
At least 200 integrated third-party tools, selection accuracy at least 90%, and average end-to-end success at least 90% in tasks requiring three or more tools. CN-A1

17. Strategic decision-making agents

266505X / 4 | Artificial intelligence | Proposed restricted
All three groups apply: at least 20 planning steps or 10 concurrent agents; adaptive decisions within 200 ms with at least 10 control units; and a standard game benchmark above 90% of humans. CN-A1

18. AI processor design

266506X / 1 | Artificial intelligence | Proposed restricted
Design meeting any listed characteristic, including 14 nm or smaller fabrication, floating-point or integer capability, advanced packaging, fast links, novel architectures or named applications. It is broader than one chip-size test. CN-A1

19. Memory chip design

266506X / 2 | Artificial intelligence | Proposed restricted
Specified 3D flash or memory design. Alternative flash tests include density, I/O speed or bonding; memory tests include 18 nm or smaller processing, vertical channels, bonding or multilayer stacking. CN-A1

20. Fast cluster connections

266507X / 1 | Artificial intelligence | Proposed restricted
Inter-chip or inter-node connections with bidirectional bandwidth at least 800 GB/s in clusters with more than 10,000 AI accelerator cards. CN-A1

21. Recovery during distributed training

266507X / 2 | Artificial intelligence | Proposed restricted
In clusters above 10,000 cards, any listed recovery or waste criterion, such as average checkpoint recovery at most 120 seconds, visible interruption at most 30 seconds or network-related wasted compute at most 5%. CN-A1

22. Efficient cluster communication

266507X / 3 | Artificial intelligence | Proposed restricted
In clusters above 10,000 cards, any listed test: linear scaling at least 98%; specified routing and balancing with latency at most 1 microsecond; or communication hiding at least 90%. CN-A1

23. Mixed-vendor cluster scheduling

266507X / 4 | Artificial intelligence | Proposed restricted
In clusters above 10,000 cards, unified scheduling or joint inference across chips from at least two vendors with a listed protocol, efficiency, recovery or geographic-scheduling property. CN-A1

24. Cell and gene therapy platforms

262704J / 1 | Biomedicine | Proposed prohibited
Chinese-owned original reusable technology platforms. The prohibition is limited to transferring ownership abroad; non-ownership licensing and specified services or single-drug work are excluded. CN-A1

25. siRNA medicine platforms

262705J / 1 | Biomedicine | Proposed prohibited
Chinese-owned original reusable siRNA drug platforms, with the same ownership-transfer boundary and stated exclusions for licensing, services and particular drug work. CN-A1

26. Specified pathogen enhancement technology

267302J / 1 | Biomedicine | Proposed prohibited
Specified artificial genetic modification or directed-passage enhancement technologies with any listed characteristic. This is a biosecurity category, not all biomedical research. CN-A1

27. New cell and gene therapy targets

262705X / 1 | Biomedicine | Proposed restricted
Chinese-owned technology to discover and validate wholly new targets not publicly disclosed worldwide through the stated routes. Restricted-biomedicine facilitation applies to specified circumstances. CN-A1

28. New siRNA targets

262706X / 1 | Biomedicine | Proposed restricted
Chinese-owned discovery and validation technology for wholly new siRNA targets, using the same undisclosed-target definition and restricted-biomedicine facilitation note. CN-A1

29. High-capacity cold hydrogenation

263003J / 1 | Solar energy | Proposed prohibited
A cold-hydrogenation fluidized-bed process with single-line trichlorosilane capacity at least 250,000 tonnes per year, and equipment-manufacturing technology. CN-A1

30. High-capacity silane fluidized beds

263003J / 2 | Solar energy | Proposed prohibited
Silane fluidized-bed processes with single-line capacity at least 20,000 tonnes per year, and equipment-manufacturing technology. CN-A1

31. Ultra-thin wafers

263004J / 1 | Solar energy | Proposed prohibited
Technology for wafers 70 micrometers thick or thinner, including associated equipment-manufacturing technology. The note includes code, underlying algorithms and drawings. CN-A1

32. Specified cell-edge treatment

263801J / 1 | Solar energy | Proposed prohibited
Solar-specific edge passivation for crystalline-silicon cells, with full-wafer edge lengths 182 to 230 mm and wafer thickness at most 140 micrometers, plus equipment-manufacturing technology. CN-A1

33. Perovskite and tandem manufacturing

263801J / 2 | Solar energy | Proposed prohibited
Four specified routes: mass-production silicon-perovskite tandem cells at least 182 mm and 30% efficient; slot coating at least 1.1 m wide; single-junction modules at least 1 m2 and 18%; or tandem modules at least 1 m2 and 28%. CN-A1

34. Large polysilicon plant engineering

263004X / 1 | Solar energy | Proposed restricted
Single-line polysilicon engineering capacity at least 30,000 tonnes per year. Engineering includes equipment and building designs and process packages. CN-A1

35. High-output reduction processes

263004X / 2 | Solar energy | Proposed restricted
Reduction processes with polysilicon output at least 800 tonnes per furnace per year, and equipment-manufacturing technology. CN-A1

36. Crystal-growing designs

263005X / 1 | Solar energy | Proposed restricted
Specified solar crystal-growing designs, including chamber diameter at least 1,600 mm, thermal-field size at least 35 inches, pulling speed at least 1.2 mm/minute or equal-diameter power at most 40 kW. CN-A1

37. Silicon-ingot machining

263005X / 2 | Solar energy | Proposed restricted
Specified half-ingot cutting/grinding technology or machining technology for ingots at least 720 mm long, plus equipment-manufacturing technology. CN-A1

38. Large or thin wafer processes

263005X / 3 | Solar energy | Proposed restricted
Specified rectangular-wafer technology with any side at least 182 mm; or listed cutting technology for wafers at most 130 micrometers thick. The thin-wafer branch has additional wire or machine tests. CN-A1

39. High-efficiency TOPCon and back-contact cells

263802X / 1 | Solar energy | Proposed restricted
Six specified process groups cover coatings, diffusion and electrodes. Several combine 182 to 230 mm wafers with thickness at most 140 micrometers and stated production rates. The high-efficiency definition also applies. CN-A1

40. Heterojunction cell processes

263802X / 2 | Solar energy | Proposed restricted
Four specified groups cover conductive films, silicon films, low-silver materials and copper electrodes. Some branches require wafers at most 110 micrometers thick and further output or material tests. Check the complete high-efficiency note. CN-A1

03. AI: what is actually covered

A model is more than a chat window

Weights are learned numbers that make a model work. Source code tells a computer how to build or run it. Checkpoints are saved stages of training. Architecture is the design. A hosted service lets us ask questions without receiving all those parts. Attachment 1 names weights, key checkpoints, underlying code, architecture, training improvement and faster model operation in its advanced-model entry. CN-A1, 266504X(2)

Three alternative tests for advanced models

Proposed entry 266504X(2) covers core technology of a large model meeting any one of three tests: training calculation of at least 10^26 floating-point operations, at least 2 trillion parameters, or globally leading ability. The last test means a top-10 public ranking at the time of export across the kinds of assessments named in the draft, including overall or specialist ability, use, safety and training efficiency. It is not confined to one popular chat leaderboard. CN-A1, 266504X(2), note 3
The parameter test and calculation test are alternatives, not a pair that must both be met. A smaller model could qualify through the ranking test. A claim that a named model crosses a threshold needs evidence about that exact version and the time of export. We do not assign legal classifications to DeepSeek or Qwen from their brand names. CN-A1

Data creates a separate boundary

Models trained on China's core data and then provided as services appear in the proposed prohibited section, 266501J. Models trained on important data and then provided as services appear in restricted entry 266504X(1). Core and important data are legal categories determined by the Data Security Law and related rules, not everyday labels for valuable data. CN-A1, 266501J and 266504X
The important-data entry has a qualified exclusion. One route concerns a trace share of at most 0.1%, with further conditions about technical capability, cleaning and preventing reproduction in outputs. Another concerns data that can lawfully be obtained publicly. A small percentage alone is not a safe conclusion. The exact structure of the Chinese note must be checked. CN-A1, 266504X, note 1

Agents, chips and the system around the model

Agents can break a request into tasks, use tools or coordinate other agents. The draft names four restricted capabilities with performance conditions. An agent used mainly in highly sensitive settings can be proposed prohibited when it meets the referenced performance criteria. General-

purpose agents are not all placed in that category merely because they can act. CN-A1, 266502J and 266505X

AI chip design and memory chip design have their own proposed restrictions. The processor-design entry uses alternative characteristics, some broadly worded, including floating-point or integer capability and specified applications. It should not be summarized as a 14 nm-only rule. The footnote offers facilitation for overseas chip fabrication; the announcement does not establish a general ban on all foreign foundry work. CN-A1, 266506X and note 2

Cluster entries concern systems with more than 10,000 AI accelerator cards. They cover fast links, recovery after failures, communication scheduling and mixed-vendor computing, subject to their listed tests. The target includes the knowledge that keeps a huge computer installation working, not only the model inside it. CN-A1, 266507X

Overseas deployment and remote access

MOFCOM says overseas local deployment of restricted model technology falls within technology-export management. An overseas cloud owned by the Chinese business may qualify for easier licensing. Its answer on restricted large models places API access, apps, web interaction, SaaS and lightweight local deployment outside large-model technology-export management. That answer should not be extended automatically to prohibited core-data technology, private weights handed to a partner, or every other law. CNQA, overseas deployment question

04. The future of open models from China

Three meanings that we must separate

Already public	Draft exclusion
First open release	Facilitated permission
Private technology	Check criteria and transfer

Already public / Draft exclusion / First open release / Facilitated permission / Private technology / Check criteria and transfer / CN-A1

A model can have downloadable weights, an open software license, and enough published training information to reproduce it. These are different properties. The Open Source Initiative's definition asks for freedoms to use, study, change and share, along with data information, code and parameters. A model described casually as open source does not automatically satisfy that definition. That definition is also not the Chinese regulator's legal test. OSI

DeepSeek's R1 repository says its code and model weights use the MIT license and identifies additional ancestry details for distilled versions. Qwen's Qwen3 repository documents downloadable models and an Apache 2.0 license. These are concrete examples of existing public releases, not proof of the status of every model from either company or of any future release. R1 QWEN

Already public and about to become public

The AI section's first footnote excludes technology already in the public domain through open source or academic publication. Its second footnote offers automatic licensing or similar facilitation for open-source exports of restricted AI technology. The distinction is meaningful: an existing public artifact and a first release of private technology need not follow the same route. CN-A1, notes 1 and 2 CNQA

Publishing a paper about a model does not necessarily publish its private weights or training code. We should identify the particular artifact and what was actually made available. This is our reading of a technology-specific exclusion. The announcement does not resolve every question about gated downloads, a partly open release or later private improvements. CN-A1

Four possible paths, not four predictions

Continued open releases. Easier licensing works as described, and firms keep releasing models because broad use is valuable. Evidence supporting this possibility is the express open-source facilitation policy. Evidence that would weaken it includes new rules narrowing that policy or documented delays. CN-A1 CNQA

More preparation before release. Teams document what they publish, which data they used and which threshold applies. This is our inference from the alternative tests and the distinction between public and newly released technology. It does not prove that releases will slow. CN-A1, 266504X

A divided product line. A company releases one model publicly while keeping a more advanced version, special dataset or agent system private. This is a business possibility, not a requirement in the announcement. It follows from controls attaching to specific technology and transfer forms. CN-A1 CNQA

More hosted access. Some providers choose services instead of transferring core technology to customers. MOFCOM's service clarification makes that plausible for restricted large models. It does not guarantee price, continued service, privacy protection or immunity from data-related prohibitions. CNQA

What would change our view

We would watch the final catalogue, definitions of public release, the practical automatic-licensing procedure, treatment of new checkpoints and derivatives, and export-time ranking methods. We would also watch access to computing hardware under US rules. No source here supports a numerical probability that Chinese open models will disappear, or a date when they will stop. We assign neither. CN-A1 EAR742 EAR744

For an overseas developer, an existing permissive license is useful evidence of reuse rights. It is not a promise of future versions or a general export-law clearance. A practical record contains the exact release, license, download date and changes made later. That is our documentation suggestion, not an added legal duty.

05. Biomedicine: platforms, targets and cooperation

The reusable engine and the individual medicine

The proposed prohibited entries include Chinese-owned original cell and gene therapy platforms and siRNA drug platforms. A platform here is reusable underlying technology that can support several drugs. The notes limit these prohibitions to overseas transfers of ownership. Licensing and contract research or production without an ownership transfer are outside that platform scope. Technology for a single drug is also distinguished from the reusable platform. CN-A1, 262704J and 262705J

The notes also exclude technology needed for development, trials, production, quality control, marketing and compliance in cooperation or business deals involving particular medicines at different stages. We should check the actual subject of a deal. A transaction described as a license can still raise questions if its substance transfers the platform's ownership. That is an inference from the ownership boundary. CN-A1

New targets have a different route

The restricted entries concern discovery and validation of wholly new targets for cell and gene therapy and siRNA drugs. The target must not have been publicly disclosed worldwide through the named routes, including patents, papers, public databases, technical materials and conferences. A promising drug is not the same thing as a wholly new undisclosed target. CN-A1, 262705X and 262706X

The biomedicine restricted-section footnote offers automatic licensing or similar facilitation for public-health emergencies, humanitarian help, joint research using overseas resources and overseas licensing. We cannot carry that footnote over to every proposed prohibited technology. CN-A1, note 3

The biosecurity entry

The pathogen entry concerns specified enhancement and directed-passage technology. We describe its legal subject at a high level rather than giving laboratory methods. It does not establish a blanket ban on medical research or ordinary medicines. CN-A1, 267302J

How the US comparison helps

The US Commerce Control List has biological agents, equipment and related technology entries, and EAR 744.4 addresses chemical and biological weapons end uses. That structure is not a

direct equivalent of China's proposed protection of reusable drug platforms. The US Data Security Program separately controls certain access to sensitive personal and government-related data. A genomics project can therefore raise data questions as well as technology questions. EAR774, Category 1 EAR744, 744.4 DOJ

An example, with limits

A company licenses rights for one named drug abroad while keeping ownership of its reusable platform. The platform notes are directly relevant, and the new-target provisions may require a separate check. We cannot treat all technology inside one contract as a single undifferentiated object. This example illustrates how to read the categories; it does not approve a contract. CN-A1

06. Solar: knowledge can travel inside a machine

Three stages of production

The solar section covers making polysilicon, turning silicon into wafers and making solar cells. Proposed prohibitions include specified high-capacity polysilicon processes, wafer technology at 70 micrometers or thinner, specified edge-passivation technology and specified perovskite or tandem-cell manufacturing. These are process and equipment boundaries, not a blanket ban on finished solar panels. CN-A1, solar prohibited section CNQA

Restricted points cover polysilicon engineering and reduction, crystal growth, machining, large or thin wafers, and specified high-efficiency cell processes. Many tests combine size, thickness, throughput or efficiency. The correct relationship between those tests matters more than a single eye-catching number. CN-A1, 263004X, 263005X and 263802X

The equipment exception to a simple story

Attachment 1's fourth footnote expressly includes key equipment carrying the relevant technology. It names categories such as specified reactors, wafer-cutting machines, crystal-growing furnaces, coating tools and cell-processing equipment. MOFCOM says the equipment provision concerns machines carrying technology that meets the control indicators; equipment not named is generally outside this announcement's requirement. CN-A1, note 4 CNQA, solar question

Selling a finished panel, sharing factory drawings and installing a production line are different activities. Installation, debugging, improvement and upgrades can transfer know-how as well as machinery. The official explanation discusses licensing procedures for covered technology transfers and temporary registration for the listed key equipment. We retain the distinction between the proposal and registration already in place. CNQA

A factory example

Suppose a supplier sends an overseas buyer an entire cell-production line with operating software and training. We identify each machine, its performance and the knowledge being delivered. We then check the equipment footnote, technology points, transfer form and current registration duty. Treating the delivery as only hardware would miss part of the announced scope. This is an explanatory example. CN-A1 CN48

Trade policy is a different layer

Tariffs change the cost of imports. Export controls regulate whether or how an outward transfer may occur. Neither word can substitute for the other. The US comparison in this book focuses on technology and AI controls; it does not turn solar tariff disputes into equivalent technology bans. A controlled manufacturing recipe can affect overseas factory plans without implying that every resulting product is controlled. This is our analysis of the technology/product distinction in the official explanation. CNQA EAR774

07. America's controls, in plain language

Four questions before an answer

In the US system we ask whether the item falls under the Export Administration Regulations, how it is classified, where it is going, and who will use it for what. US-origin items and some foreign-made items can be covered. The foreign-made rules have specific conditions; a vague connection to America does not by itself settle jurisdiction. EAR734, 734.3 and 734.9 EAR744

The Commerce Control List assigns classification codes. Advanced computing chips are addressed in entries including 3A090. End-user and end-use rules can create requirements beyond the basic destination table. The Entity List is one part of this system. Inclusion and license conditions must be checked for the actual parties, rather than inferred from a country label. EAR774 EAR744, 744.11 and 744.23

The January 2026 chip change

The 15 January final rule permits case-by-case review for qualifying exports from the US to end users in China or Macau. The stated performance limits are total processing performance below 21,000 and total DRAM bandwidth below 6,500 GB/s. H200 and MI325X are examples, not a substitute for classifying the exact product. US26 EAR742, 742.6(b)(10)

Conditions include adequate US supply, no diversion of relevant foundry capacity from US users, a limit relating shipments to China and Macau to US shipments, customer checks, security measures and independent testing in the US. Approval remains discretionary. The rule does not give the same favorable policy to every reexport or shipment to a China-linked company in another country. US26, sections I and IV

The AI Diffusion Rule: written law and enforcement

The January 2025 framework introduced controls on certain advanced model weights and broader computing exports. Current BIS-published text still includes 4E091 for model parameters trained using at least 10^26 operations. Its notes exclude published parameters and provide qualified treatment for derivatives and models below an open-model capability reference. The full notes matter. US25 EAR774, 4E091

On 13 May 2025, BIS announced non-enforcement and a planned rescission. On 12 May 2026, GAO said the rule remained legally in effect pending formal rulemaking and concluded that the blanket non-enforcement policy itself required submission under the Congressional Review Act. The current text checked for this book still contains the model-weight entry. We therefore do not describe the rule as simply repealed, or portray its licensing provisions as straightforwardly

enforced. A press release, a rulebook and a legal oversight decision answer different questions. USNE GAO26 EAR774

Public information has an important place

EAR 734.7 generally treats qualifying published, unclassified technology and software as outside the EAR, with stated exceptions. EAR 734.8 addresses results of fundamental research intended for publication. Published model parameters are expressly addressed in 4E091. These provisions do not make every private checkpoint, encryption tool or restricted release exempt just because a related paper exists. EAR734 EAR774

Nearby rules are not interchangeable

Treasury's outbound investment program covers certain US investments involving semiconductors, quantum technology and AI in relation to China, Hong Kong and Macau. The 2024 final rule took effect on 2 January 2025. Treasury issued a further proposal on 22 October 2026, with comments due on 7 December. A new proposal is not automatically a new final investment rule. TREAS

The Justice Department's Data Security Program concerns specified access to sensitive data. These programs can overlap with an AI project, but neither is the same as an export license for a chip or a model. We label them separately to avoid double counting or claiming that one approval answers every question. DOJ TREAS

08. What the comparison tells us

Compare the gates, not the slogans

China's Announcement 48 focuses on technology leaving China, including ownership and control changes. The US framework focuses on covered items, destinations, recipients and uses, with a reach to some foreign-made items. Both can affect an AI project at different points. These are summaries of the cited systems, not claims that either country controls all technology worldwide. CN48 EAR734 EAR742 EAR744

Question	China / Announcement 48	United States / EAR
Main gate	Technology transfer and ownership or control changes	Covered items, destination, user and use
Public AI	Draft exclusion for already public technology	Published-information provisions and 4E091 notes
New open release	Facilitation proposed for restricted AI exports	Check publication provisions and model-rule status
Hardware	Chip-design knowledge and specified solar equipment	Advanced chips and specified manufacturing items
Status caution	Registration now; proposed new catalogue categories	Chip controls; model text with non-enforcement dispute

CN48 CN-A1 CNQA EAR734 EAR742 EAR774 GAO26

The same project can meet both systems

Consider a hypothetical Chinese developer that uses covered US chips at an overseas computing center and later licenses private model technology to a foreign partner. US rules may matter to the chips, users and uses. China's proposed technology categories and current registration may matter to the model transfer. A lawful path under one system does not establish a lawful path under the other. This is an illustrative interaction, not a claim about a real company. CN48 CN-A1 EAR734 EAR744

What we can and cannot infer

Controls could raise documentation costs, encourage hosted services or influence where firms build factories and computing centers. Those are plausible mechanisms. The announcement alone does not quantify their effect on prices, jobs, investment, clean-energy adoption or AI performance. We make no numerical forecast. Evidence for an outcome would require observed transactions, implementation data and a comparison with what would have happened without the controls.

A similar number in two rules can hide different meanings. China's 10^26 test is one of several alternative advanced-model criteria. The US 4E091 entry has its own operations definition, exclusions and enforcement complication. We cannot call the systems identical because one threshold looks similar. CN-A1, 266504X EAR774, 4E091 GAO26

09. Questions we can ask now

Does this end downloads of Chinese open models?

The announcement does not say that. The draft excludes AI technology already public through open source or papers, and it offers facilitation for open-source exports of restricted AI technology. Treatment of a particular new release still depends on its facts and the implemented procedure. CN-A1, notes 1 and 2

Is using a Chinese chatbot an export of the model?

MOFCOM's answer on restricted large models distinguishes remote services from transferring model technology. That is a useful boundary, not a universal exemption for data, prohibited technologies or other activities bundled into a service. CNQA

Is buying an AI company abroad just a financial transaction?

The temporary registration list includes an overseas entity becoming the owner or actual controller of domestic technology. We must therefore ask about control of the technology as well as the corporate deal. The announcement does not say every foreign investment is prohibited. CN48, section III(2)

Can a solar machine or a drug license be treated as ordinary goods?

Specified solar equipment is expressly included when it carries covered technology. Biomedicine has specific ownership-transfer exclusions and licensing facilitation. We identify what actually changes hands before choosing a category. CN-A1, notes 3 and 4 and biomedicine notes

What should a useful evidence record contain?

Our suggested record names the technology version, relevant code and point, measured performance, public/private artifacts, data categories, transfer form, recipient, destination and dates. It attaches the original source and explains any exclusion relied on. The registration form lists its own required fields; our record is a reading aid, not a substitute form. CN-A2

What can still change?

The final catalogue, detailed procedures, authoritative interpretations and later US rulemaking can change the answer. This edition fixes its research date so that future readers can see its limits. We should recheck a live rule before using this book for a real decision.

An existing public model

Already public AI technology is excluded in the draft. Identify the exact artifact, release date and license. A related paper does not necessarily disclose private weights. CN-A1, note 1

A first open release

The draft offers automatic licensing or similar facilitation for open-source exports of restricted AI technology. A first release is different from an already public artifact. The implemented process still matters. CN-A1, note 2

Private weights for a partner

Check the exact technology and alternative thresholds in 266504X. Overseas licensing can trigger current registration; do not assume a software license is government permission. CN48 CN-A1

A hosted model service

The official answer distinguishes API, web, apps, SaaS and lightweight local deployment for restricted large models. It is not a universal exception for prohibited core-data technology or other laws. CNQA

A change of ownership

Registration includes a domestic technology owner or actual controller becoming an overseas entity. Examine control of the technology, not just the physical movement of files. CN48, III(2)

A solar production line

Check each listed equipment category and the technology indicators, plus software, installation and training. Covered key equipment is included in temporary registration. CN-A1, note 4 CNQA

10. Sources, method and small glossary

How this book was researched

We read the official announcement, both downloaded WPS attachments and the official questions and answers. We compared US Federal Register rules with BIS's current published regulatory text, BIS's non-enforcement announcement and GAO's May 2026 decision. We checked Treasury and Justice Department sources for adjacent programs, and model publishers for concrete open-release examples. Research was checked on 2 November 2026.

We use short source IDs beside claims. The list below gives the original link, date and a location within the source. The atlas identifies each point by code. The research folder preserves downloaded bytes, hashes and download limitations. All 40 summaries are editorial paraphrases. They do not replace the Chinese attachment or the current US rules.

The official files are WPS documents. Our PDF reading copies were converted locally with LibreOffice. We also provide a PDF print of the official announcement page. Conversion may change pagination. The original WPS bytes and source links allow comparison. We do not relabel the unrelated user-supplied organoid PDF as Announcement 48.

Translation and uncertainty

All seven editions cover the same chapters and 40-point atlas. Translations are editorial reading aids and have not received independent legal or native-language professional review. The original legal texts prevail. English words inside catalogue codes, product names and source titles remain where needed for verification. Diagrams simplify relationships and show no invented quantitative forecasts.

Statements about what a rule says carry sources. Hypothetical examples and possible futures are explicitly marked. We do not claim that a company will stop publishing models, that an individual transaction is approved, or that our search excludes the existence of every later legal document.

A small glossary

Export: a transfer covered by a rule. It can involve knowledge or rights rather than a physical shipment.

Weights: the learned numerical settings of an AI model.

Training: the process that adjusts those settings using data and computation.

Inference: using a trained model to produce an answer.

Agent: a system that can plan steps and use tools to work toward a task.

API: a way for one program to ask another program for a service.

Open weights: model settings made available for download; the phrase alone says little about training code or data.
License: either a government permission or a copyright permission, depending on context. We always distinguish the two.
Proposed prohibited / restricted: a category in the consultation draft, not a claim that all its new final rules already operate.

Sources

CN48

MOFCOM and MOST, Announcement 48
2026-10-16 | Sections I, II and III

CN-A1

Announcement 48, attachment 1 (official WPS original)
2026-10-16 | AI, biomedicine, solar and four footnotes

CN-A2

Announcement 48, attachment 2 (official WPS original)
2026-10-16 | Registration form fields

CNQA

MOFCOM spokesperson, questions and answers
2026-10-16 | Questions on AI, biomedicine, solar and registration

US26

BIS, semiconductor license review final rule
2026-01-15 | 91 FR 1684; sections I and IV; 742.6(b)(10)
PDF

US25

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2025-01-15 | 90 FR 4544; ECCN 4E091 and published weights
PDF

USNE

BIS, announcement of rescission and non-enforcement
2025-05-13 | Non-enforcement directive; planned formal rulemaking

GAO26

GAO, decision B-337935
2026-05-12 | Commerce Response; CRA Exceptions; Conclusion

EAR734

BIS, current EAR Part 734
2026-11-01 | 734.3, 734.7, 734.8 and 734.9

EAR742

BIS, current EAR Part 742
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EAR744

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2026-11-01 | 744.1, 744.4, 744.11 and 744.23

EAR774

BIS, current Commerce Control List
2026-11-01 | Categories 1, 3 and 4; 3A090 and 4E091

TREAS

US Treasury, Outbound Investment Security Program
2026-10-22 | 2024 final rule and October 2026 proposed rule

DOJ

US Justice Department, Data Security Program
2026-11-02 | Overview and program rules

OSI

Open Source Initiative, Open Source AI Definition 1.0
2024-10-28 | Freedoms; data information, code and parameters

R1

DeepSeek, R1 repository and license
2025-01-20 | License; released weights; distilled variants

QWEN

Qwen team, Qwen3 repository
2025-04-29 | Release history, model files and Apache 2.0 license

GD

GuideDog repository and publishing instructions
2026-11-02 | README language list and HTML/PDF build commands